ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

# INTRODUCTION TO ROBOTICS

## PAPER 1 · PRACTICE A · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 3 Introduction to Robotics paper 1 · practice a.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

## SECTION A

## QUESTION 1

**1.1** C ✓
**1.2** A ✓
**1.3** B ✓
**1.4** D ✓
**1.5** A ✓
**1.6** C ✓
**1.7** C ✓
**1.8** A ✓
**1.9** D ✓
**1.10** B ✓

(10 × 1) [10]

## QUESTION 2

**2.1** True ✓
**2.2** True ✓
**2.3** True ✓
**2.4** False ✓
**2.5** True ✓
**2.6** True ✓
**2.7** False ✓
**2.8** True ✓
**2.9** False ✓
**2.10** True ✓

(10 × 1) [10]

## QUESTION 3

**3.1** A ✓
**3.2** F ✓
**3.3** G ✓
**3.4** I ✓
**3.5** B ✓
**3.6** C ✓
**3.7** H ✓

**3.8** J ✓

**3.9** D ✓

**3.10** E ✓

(10 × 1) [10]

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 4

**4.1** (a) Cartesian – three linear joints ✓, rectangular box-shaped envelope ✓ (b) cylindrical – one rotary base joint and two linear joints ✓, cylinder-shaped envelope ✓ (c) articulated – rotary joints (usually six) ✓, roughly spherical / irregular envelope ✓ (6)

**4.2** (a) SCARA or delta robot ✓ – very fast, accurate horizontal pick-and-place ✓ (b) articulated (6-axis) robot ✓ – can reach around and into the car body at any angle ✓ (4)

**4.3** Any THREE named ✓ and explained ✓: world (base) frame – fixed to the robot base / cell, used for general positions; joint coordinates – each axis moved individually by angle; tool frame – based on the tool centre point, moves with the tool, used to approach parts along the tool direction; user (work-object) frame – defined on a fixture or table, so a program can be moved by redefining the frame (6)

**4.4** Accuracy – how close the robot gets to the commanded target position ✓✓; repeatability – how close it returns to the same position each time it repeats the movement ✓✓ (a robot can be repeatable without being accurate). (4)

**[20]**

## QUESTION 5

**5.1.1** $x = 30\cos 30° + 20\cos 90° \checkmark = 25{,}98 + 0 \checkmark = 25{,}98\text{ cm} \checkmark$; $y = 30\sin 30° + 20\sin 90° \checkmark = 15 + 20 \checkmark = 35\text{ cm} \checkmark$ (6)

**5.1.2** $30 + 20 = 50\text{ cm} \checkmark$ (1)

**5.2** There may be more than one set of joint angles (or none) for a position ✓, and the equations are non-linear and harder to solve ✓. (2)

**5.3** Point-to-point: only the start and end positions matter; the path between is not controlled ✓, e.g. spot welding, pick and place ✓; continuous path: the tool follows a controlled path all the way ✓, e.g. arc welding, painting, gluing ✓ (4)

**[13]**

## QUESTION 6

**6.1** $V_{avg} = 0{,}75 \times 12 \checkmark = 9\text{ V} \checkmark$ (2)

**6.2** (a) $90° \div 1{,}8° \checkmark = 50\text{ steps} \checkmark$ (b) $1\,000 \div 200 = 5\text{ revolutions} \checkmark$; $5 \times 2\text{ mm} \checkmark = 10\text{ mm} \checkmark$ (5)

**6.3** DC motor: speed control by voltage/PWM, position needs extra feedback ✓ – wheels of a mobile robot ✓; servo motor: built-in feedback, moves to a set angle ✓ – grippers, steering, small arm joints ✓; stepper motor: moves in exact steps without feedback (open loop) ✓ – 3D printers, CNC axes, Cartesian robots ✓ (6)

**6.4** A slotted / marked disc on the shaft turns between a light source and a sensor ✓; each slot produces a pulse ✓; the controller counts the pulses (and their sequence) to calculate the angle, speed and direction ✓. (3)

**6.5** It sends a sound pulse and measures the time until the echo returns; the sound travels to the object and back ✓. $d = (v \times t) \div 2$ ✓ $= (343 \times 0{,}0029) \div 2$ ✓ $= 0{,}50$ m $(49{,}74$ cm$)$ ✓ (4)

**[20]**

## QUESTION 7

**7.1** Forward: both wheels turn forward at the same speed ✓; turn left: the right wheel turns faster than the left wheel ✓; spin on the spot: wheels turn at the same speed in opposite directions ✓ (3)

**7.2** Any THREE (method ✓ + advantage ✓): teach pendant – simple and intuitive, positions taught on the real robot; offline programming / simulation – no production downtime, test safely; lead-through (hand guiding) – fast for complex paths such as painting, no coding needed; text-based programming (e.g. RAPID, Python) – precise, logic and calculations possible (6)

**7.3** Any FOUR with purpose ✓✓✓✓: LiDAR – map the corridor and avoid obstacles; ultrasonic sensors – detect nearby objects / people; bump sensors – stop on contact; wheel encoders – odometry; camera – recognise doors, signs, people; IMU – orientation; cliff sensors – detect stairs (4)

**7.4** Any FOUR with a task ✓✓✓✓: two-finger gripper – picking parts; vacuum gripper – lifting boxes or glass; magnetic gripper – steel sheets; welding torch – welding; spray gun – painting; screwdriver – assembly (4)

**[17]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**